

Math 8 End of Q4 Test

Mary, Star of the Sea School · Dr. Park

Name: _____ Class: _____ Date: _____

Directions: Answer all 20 questions in the space provided.

1. $V = 20 \text{ V}$, $R = 5 \Omega$. I ?

Answer: _____

2. $I = 3 \text{ A}$, $R = 10 \Omega$. V ?

Answer: _____

3. $v = f\lambda$: $f = 10 \text{ Hz}$, $\lambda = 3 \text{ m}$. v ?

Answer: _____

4. $v = 300 \text{ m/s}$, $f = 50 \text{ Hz}$. λ ?

Answer: _____

5. If λ halves, f (constant v) ____?

Answer: _____

6. Kepler: r triples. T changes by factor of $\sqrt{27} \approx ?$

Answer: _____

7. Earth $T = 1 \text{ yr}$. Planet at $r = 9 \text{ AU}$. T (yrs)?

Answer: _____

8. Cylinder $r = 4$, $h = 6$, $\pi = 3.14$. V ?

Answer: _____

9. Cone $r = 6$, $h = 4$, $\pi = 3.14$. V ?

Answer: _____

10. Sphere $r = 6$, $\pi = 3.14$. V ?

Answer: _____

11. Two-way table: 25 in row A, 35 in row B. Total?

Answer: _____

12. Table shows 80 total, 30 in cat X. Cat Y?

Answer: _____

13. $P = VI$: $V = 20$, $I = 4$. Power?

Answer: _____

14. 650 nm to meters ($1 \text{ nm} = 10^{-9} \text{ m}$)?

Answer: _____

15. Pyramid $V = \frac{1}{3}Bh$. $B = 4 \times 5$, $h = 12$. V ?

Answer: _____

16. Cone $r = 5$, $h = 12$, $\pi = 3.14$. V ?

Answer: _____

17. Kepler: closer orbit = shorter period. True?

Answer: _____

18. 60 W bulb for 5 hrs. Energy?

Answer: _____

19. Two-way tables show what kind of variables?

Answer: _____

20. Cube side 8. Volume?

Answer: _____